

Practice Tutorial: Detailed Solutions

INF1004 Mathematics II

SIT

Solution 1: Gaussian Elimination (Part 1)

Question: Determine the solution to the system.

1. Represent as an Augmented Matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & -2 & 1 \\ -3 & -3 & -3 & -3 \end{array} \right]$$

2. Elementary Row Operations:

- $R_2 \rightarrow R_2 - R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -3 & 0 \\ -3 & -3 & -3 & -3 \end{array} \right]$$

- $R_3 \rightarrow R_3 + 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Solution 1: Gaussian Elimination (Part 2)

3. Interpret the Row-Echelon Form: The third row $[0, 0, 0|0]$ indicates that the equations are dependent, leading to **infinitely many solutions**.

4. Solve for variables: From Row 2:

$$-2y - 3z = 0 \implies y = -\frac{3}{2}z$$

From Row 1:

$$x + y + z = 1$$

Substitute $y = -\frac{3}{2}z$:

$$x + \left(-\frac{3}{2}z\right) + z = 1 \implies x - \frac{1}{2}z = 1$$

$$x = 1 + \frac{1}{2}z$$

Solution 1: Gaussian Elimination (Part 3)

5. Parametric Solution: Let z be a free parameter t . The general solution is:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 + \frac{1}{2}t \\ -\frac{3}{2}t \\ t \end{pmatrix}$$

Verification with previous result: If we isolate t from $y = -\frac{3}{2}t$, we get $t = -\frac{2}{3}y$. If we isolate t from $x = 1 + \frac{1}{2}t$, we get $t = 2x - 2$.

Final Answer:

$$z = -\frac{2}{3}y = 2x - 2$$

Solution 2: Unit Vector of a Line

Question: Find unit vector $\hat{v}$ for $z = -y = 2x + 1$. **Step-by-Step Analysis:**

- Set to parameter t : $2x + 1 = t$, $-y = t$, $z = t$.

- Express in t : $x = \frac{1}{2}t - \frac{1}{2}$, $y = -t$, $z = t$.

- Direction vector $\vec{v}_{raw} = \begin{pmatrix} 1/2 \\ -1 \\ 1 \end{pmatrix}$.

- Scale for integers: $\vec{v} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.

- Magnitude: $|\vec{v}| = \sqrt{1^2 + (-2)^2 + 2^2} = 3$.

- Normalize: $\hat{v} = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.

Final Answer: $\hat{v} = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.

Solution 3: Eigendecomposition (Part 1)

Question: Determine the eigendecomposition of A .

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Analysis:

- **Eigenvalues λ :** Solve $\det(A - \lambda I) = 0$.
- $(1 - \lambda)((2 - \lambda)^2 - 1) = 0 \implies \lambda_1 = 1, \lambda_2 = 1, \lambda_3 = 3$.
- **Eigenvectors:** Solve $(A - \lambda I)\vec{x} = 0$.
- For $\lambda = 1$: $\vec{p}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{p}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$.
- For $\lambda = 3$: $\vec{p}_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

Solution 3: Eigendecomposition (Part 2)

Final Answer: $A = PDP^{-1}$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$$

SVD of Matrix B ($B^T B = A$):

- Since $B^T B = A$, the singular values of B are $\sqrt{\lambda_A}$.
- $\sigma_1 = \sqrt{1} = 1, \sigma_2 = \sqrt{1} = 1, \sigma_3 = \sqrt{3}$.
- $\Sigma_B = \text{diag}(1, 1, \sqrt{3})$.

Solution 4: Upper Diagonal System

Question: Compute the solution vector (x, y, z) . **Back Substitution**

Steps:

① $5z = 34 \implies z = 6.8.$

② $6y - 13(6.8) = 18 \implies 6y - 88.4 = 18.$

③ $6y = 106.4 \implies y \approx 17.733.$

④ $8x - 11(17.733) + 27(6.8) = 31.$

⑤ $8x - 195.066 + 183.6 = 31.$

⑥ $8x = 42.466 \implies x \approx 5.308.$

Final Answer: $(x, y, z) = (5.3083, 17.7333, 6.8).$

Solution 5: Orthogonal Unit Vector

Question: Find unit vector orthogonal to $(3, -7, 10)$ and $(-2, 1, -4)$ with $x > 0$. **Analysis:**

- Cross product for direction: $\vec{u} \times \vec{w} = (18, -8, -11)$.
- Since $18 > 0$, we use this orientation.
- Magnitude: $\sqrt{18^2 + (-8)^2 + (-11)^2} = \sqrt{509} \approx 22.56$.
- Normalize components by 22.56.

Final Answer: $(0.80, -0.35, -0.49)$.

Solution 6: Eigenvector for $c = 5$

Question: Find eigenvector for $\lambda = 5$ of matrix A .

$$A = \begin{pmatrix} -0.5 & -2 & 2.5 \\ 1.5 & -1 & -1.5 \\ -5.5 & -2 & 7.5 \end{pmatrix}$$

Step-by-Step:

- Solve $(A - 5I)\vec{v} = 0$: $\begin{pmatrix} -5.5 & -2 & 2.5 \\ 1.5 & -6 & -1.5 \\ -5.5 & -2 & 2.5 \end{pmatrix} \vec{x} = 0$.
- From $R2$: $x - 4y - z = 0 \implies x = 4y + z$.
- From $R1$: $-5.5(4y + z) - 2y + 2.5z = 0$.
- Simplify: $-24y - 3z = 0 \implies z = -8y$.
- Let $y = -2 \implies z = 16$ and $x = 8$.

Final Answer: Eigenvector is $(8, -2, 16)$.

Solution 7: Eigenvalues of S

Question: Find eigenvalues of $S = \begin{pmatrix} -4 & 2 \\ 3 & 6 \end{pmatrix}$. **Step-by-Step:**

- Solve $\det(S - \lambda I) = 0$.
- $(-4 - \lambda)(6 - \lambda) - 6 = 0$.
- $\lambda^2 - 2\lambda - 24 - 6 = 0 \implies \lambda^2 - 2\lambda - 30 = 0$.
- Quadratic formula: $\lambda = \frac{2 \pm \sqrt{4 - 4(1)(-30)}}{2}$.
- $\lambda = \frac{2 \pm \sqrt{124}}{2} = 1 \pm \sqrt{31}$.

Final Answer: $\lambda = 1 \pm \sqrt{31}$.